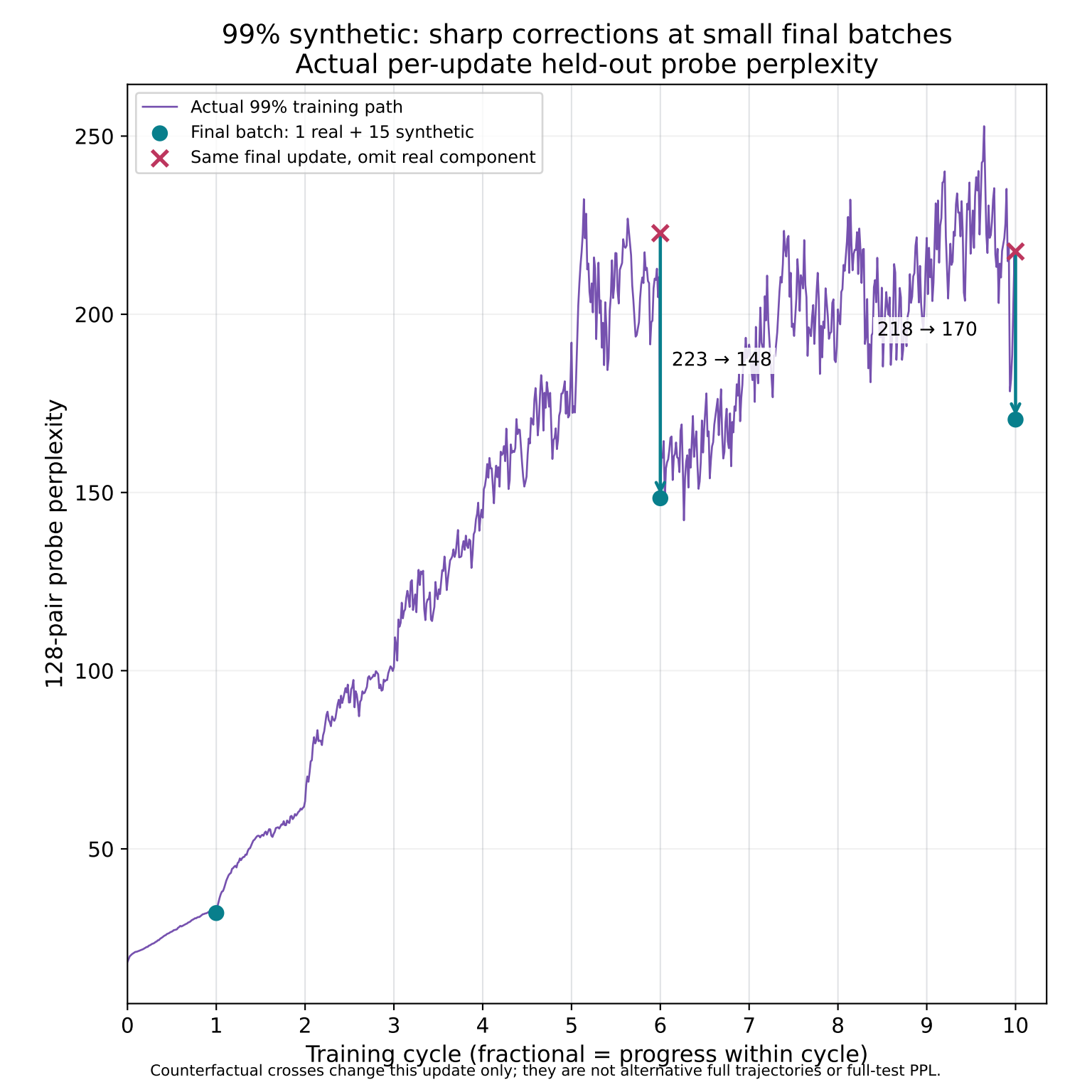
99% synthetic: sharp corrections at small final batches
Actual per-update held-out probe perplexity
Actual 99% training path
Final batch: 1 real + 15 synthetic
Same final update, omit real component
223 → 148
218 → 170
128-pair probe perplexity
Training cycle (fractional = progress within cycle)
Counterfactual crosses change this update only; they are not alternative full trajectories or full-test PPL.